Dark Matter — an Illusion

Why Galaxies Work Without Invisible Mass

Fundamental Fractal Geometric Field Theory (FFGFT) — Galactic Physics

A Popular Science Overview

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Chapter 1

The Mystery of Flat Curves

It was a simple measurement with an uncomfortable answer.

In the 1970s, Vera Rubin observed something that physics has not let go of since: the stars at the edge of a galaxy move too fast. Far too fast. According to the law of gravity — the same law that keeps planets orbiting the Sun — stars far from the centre should move more slowly than those closer in. Just as Neptune orbits the Sun more slowly than Earth does.

But they do not.

The rotation curves of galaxies — the curve showing how fast stars at different distances from the centre orbit — flatten out at the edges. The velocity stays constant as far as one can measure. It is as if all planets in our solar system orbited the Sun at the same speed, no matter how far away they are.

The standard explanation is: there must be more mass than we see. An invisible, lightless mass — dark matter — must envelop the galaxies and strengthen the gravitational pull. Without it, the stars at the edge would fly away.

This explanation sounds reasonable. But it comes at a price: dark matter has never been directly detected. Not in particle accelerators, not in underground detectors, not through any other experiment. We postulate 26 % of the entire universe to consist of a substance we do not know, cannot see, and cannot grasp.

This book asks a different question: What if galaxies are not misbehaving — but our physics simply stops working at their edges?

FFGFT — Fundamental Fractal Geometric Field Theory — gives a precise answer. It needs no new particles. It needs no invisible mass. It needs only a single number: $\xi \approx \frac{4}{3} \times 10^{-4}$.

Chapter 2

What Newton Really Says — and Where He Stops

Newton's law of gravity is one of the most reliable formulas in physics. On Earth, in the solar system, in computing satellite trajectories — it works with extraordinary precision.

But it carries a hidden assumption: that mass and inertia always bear the same relationship to each other under all circumstances. Inertia is what prevents a body from changing its velocity — the greater the inertia, the more force is needed to accelerate it.

Newton wrote: $F = m \cdot a$. Force equals mass times acceleration. This seems self-evident. But what if this relationship only holds for large accelerations?

In everyday life, accelerations are large. A falling object on Earth experiences $g \approx 10$ m/s^2. A satellite in Earth orbit perhaps a thousandth of that. But a star at the outermost edge of a galaxy experiences an acceleration of only $a \sim 10^{-10}$ m/s^2 — one ten-billionth of Earth's surface gravity.

Have we ever tested whether Newton's law holds in this regime?

The honest answer is: no. In the laboratory, in the solar system, on Earth — we have confirmed it thousands of times. In the galactic deep-field regime, we have never tested it directly.

FFGFT says: different physics applies there. Not because a new force appears, but because inertia itself becomes nonlinear.

Chapter 3

The Fundamental Relation: Time and Mass as Reciprocals

The core of FFGFT is a single formula:

$$\tilde{T} \cdot m = 1.$$

Intrinsic time and mass are reciprocals of each other. Where mass is large, time passes slowly. Where mass is small, time passes quickly. This is not a metaphor — it is a geometric property of the field.

From this relation it follows: a body with very small mass inhabits a time regime that differs fundamentally from that of a heavy body. And since inertia is connected to mass and time, inertia itself changes.

This sounds abstract. But it has a very concrete consequence for galaxies.

At the edge of a galaxy, the mass of visible matter is low, and the acceleration is very small. Both effects appear together: small mass and small acceleration. FFGFT says that in this regime, Newton's formula $F = ma$ no longer holds exactly. Inertia becomes smaller than Newton predicts — stars respond to gravity more sensitively than expected.

This is the effect we see in rotation curves — without dark matter.

Chapter 4

The Transition Scale a_0 — the Same Scale as H_0

For a theory to be scientific, it must predict precisely where an effect begins. FFGFT does this.

There is an acceleration scale a_0 below which the Newtonian regime ends and the galactic regime begins. This scale is not a separate number fitted to galaxies. It follows from ξ together with the same calibration that also fixes the Hubble constant:

$$a_0 = \frac{c^2 \xi^{10}}{4\,\bar{\lambda}_e} \approx 1.03 \times 10^{-10}\ \text{m/s}^2.$$

Here $\bar{\lambda}_e = \hbar/(m_e c)$ is the reduced Compton wavelength of the electron — a known, measurable quantity. The factor ξ^{10} carries the entire scale structure: ten powers of the fundamental constant connect the quantum scale of the electron to the scale of galactic acceleration.

This is remarkable. This value of a_0 had been known empirically in astrophysics for decades — Mordehai Milgrom determined it from observations in 1983, without being able to derive it, and he already noticed that it is of the order of cH_0. FFGFT supplies exactly this relation: $a_0 = cH_0/(2\pi)$. The exponent 10 in ξ^{10} is calibrated to the Hubble constant; once that is fixed, a_0 follows without any further parameter.

Below a_0, the modified force law applies:

$$a = \sqrt{a_N^2 + a_N a_0},$$

where a_N is the Newtonian acceleration from the visible mass. For large accelerations ($a_N \gg a_0$) this reduces to Newton: $a \to a_N$. For small accelerations ($a_N \ll a_0$):

$$a \to \sqrt{a_N a_0} \qquad \Longrightarrow \qquad v^4 = G M_b a_0.$$

This is the *baryonic Tully-Fisher relation* — one of the most robust empirical rules in galactic physics, observed for forty years, yet never derived from first principles by the standard model. FFGFT derives it.

Chapter 5

The Test: DDO 154 and the Milky Way

A theory must be measured against observations. Three galaxies of different types are particularly instructive.

DDO 154 — the cleanest test. This dwarf galaxy is gas-dominated: almost all of its visible mass consists of hydrogen gas, whose quantity is known very precisely from radio astronomy. There is little uncertainty about the visible mass — this makes DDO 154 the ideal test case.

The Newtonian prediction for the outer rotation velocity is about 18.5 km/s. The observation shows 47.0 km/s — more than twice as large.

The FFGFT prediction from the modified force law: 47.1 km/s.

The ratio of prediction to observation: 1.002 — agreement to two parts per thousand.

NGC 3198 — a spiral galaxy. Here the baryonic mass is less precisely known (uncertainty $\sim 30\,\%$). The Newtonian expectation: 50.4 km/s, observed: 150 km/s. FFGFT predicts 121 km/s, about 19 % too low. The mass uncertainty cannot explain this: in the galactic regime the velocity grows only with the fourth root of the mass, so $\pm 30\,\%$ in mass changes it by only about 7 to $8\,\%$. Closing the gap would require roughly 2.4 times the visible mass.

The Milky Way. Newton expects 122 km/s in the outer disc, measurements give around 220 km/s, FFGFT predicts 180 km/s, about 18 % too low. Here too about 2.2 times the mass would be

needed. For spiral galaxies the test is therefore open (Doc. 308); the clean test case remains DDO 154.

Dark matter explains these curves too — but only by fitting a separate halo profile for each galaxy. Those are free parameters, one per galaxy. FFGFT fits nothing to individual galaxies: a_0 follows from the same calibration as H_0 and is the same for all galaxies.

Chapter 6

The Bullet Cluster — the Seemingly Strongest Argument

Anyone who argues against dark matter eventually encounters the same objection: the Bullet Cluster.

In 2006, Douglas Clowe and colleagues published an observation considered decisive evidence for dark matter. Two galaxy clusters had collided. The hot gas — by far the largest visible mass component — was slowed by the collision and left behind. The galaxies themselves passed through.

From gravitational lensing measurements, the true location of the mass was determined: not at the gas, but at the galaxies. The gas was slowed; the mass was not. Therefore, a collision-free invisible mass must be present — dark matter.

This is a compelling argument, and FFGFT cannot yet answer it.

The modified force law acts on all mass — visible and interacting alike. What matters, therefore, is where the visible mass sits. In the Bullet Cluster the hot X-ray gas is the dominant visible component: it carries roughly five to ten times more mass than the stars of the galaxies, about 85 % of the ordinary matter. During the collision the gas was slowed by ram pressure, while the galaxies passed through.

A simulation of the collision geometry with the parameters of Clowe et al. 2006 shows for FFGFT: with the observed masses of gas and stars, the strongest lensing effect lies at the gas, about

240 kpc from the galaxies. It is observed, however, at the galaxies. The peak moves to them only if the galaxies are given a multiple of their visible mass — and precisely this additional mass is what is called dark matter. A test with it would be circular.

For FFGFT the Bullet Cluster is therefore an open test, not a passed one (Doc. 308). At present it is the strongest argument against any explanation without additional mass; similar approaches such as MOND face the same difficulty. Whether the nonlinear inertia in a collision, the time course of the event or the fractal geometry can account for the difference has not been shown.

Chapter 7

Dark Energy — a Construct of the Wrong Interpretation

Alongside dark matter, the standard model knows a second unknown: dark energy. It is supposed to drive the accelerated expansion of the universe and is said to account for 69 % of the entire universe.

Together, the two unknown components amount to 95 % of the universe. Only 5 % is ordinary matter that we know and understand.

FFGFT needs neither dark matter nor dark energy — but for different reasons.

Dark matter, as shown, is not a substance: it is a regime in which inertia becomes nonlinear. The effects are real — rotation curves are real, gravitational lensing is real — but they require no new ingredient.

Dark energy disappears for a different reason: there is no expansion to accelerate. In FFGFT the universe is static. The redshift of distant galaxies arises not because space is stretching, but because clocks and rulers run along with the mass; the path lengthening in the ξ-field describes the same redshift in this reading:

$$1 + z = e^{\xi x}.$$

For small distances this is indistinguishable from a Doppler effect. The reading in which clocks and rulers run along is equivalent to expansion: supernova light curves, the background temperature and the Tolman test cannot distinguish between the two. An energy loss of the photon alone, by contrast, would fail all three (Doc. 392).

The simple exponential form carries the distance measurements up to about $z = 1.5$; at $z = 2.33$ it deviates clearly, and a distance law that also fits there is still open.

The cosmological constant Λ — the mathematical shorthand for dark energy — has no referent in a static universe. It is a bookkeeping quantity of the expansion interpretation, not a physical object.

Chapter 8

What the Data Can Actually Decide

A fair question: if FFGFT can explain the same observations as the standard model — why switch?

The answer lies in the structure of the theories, not in individual data points.

The standard cosmological model (ΛCDM) describes cosmology with six free parameters, all empirically determined: Hubble constant, baryon density, dark matter density, spectral index, fluctuation amplitude, optical depth. The particle physics standard model adds 19 more. In total, over 25 numbers, none derived from first principles.

FFGFT has a single parameter: $\xi = \frac{4}{3} \times 10^{-4}$. From it follow:

- the gravitational constant G,
- the form of the Hubble constant, $H_0 = 66.82$ km/s/Mpc (the exponent 10 is calibrated to H_0),
- the acceleration scale $a_0 = cH_0/(2\pi) = 1.03 \times 10^{-10}$ m/s^2 without any further parameter,
- the CMB temperature $T_{\mathrm{CMB}} = 2.725$ K,
- the baryonic Tully-Fisher relation,
- the lepton mass hierarchy.

The same ξ anchors five different sectors of physics — particle physics, cosmology, galactic dynamics, gravitational lensing, quantum mechanics. That is not calibration: it is cross-anchoring.

The cosmological constant Λ, by contrast, works nowhere outside its own fit. That is the difference between a theory and a bookkeeping entry.

Where can one decide today? The classical pillars — supernovae, baryonic acoustic oscillations, CMB spectrum — are degenerate: both models produce the same mathematical curves. But there are observations that discriminate:

- **Wavelength dependence of redshift:** FFGFT predicts strictly achromatic redshift. Any demonstrated wavelength dependence would falsify the model.
- **Direct dark matter detection:** Any direct detection of a dark matter particle — in accelerators, detectors, or cosmic sources — would show that a new substance genuinely exists. A first candidate has existed since September 2026 (see below).
- **Euclid and DESI:** The new galaxy surveys will measure the growth of cosmic structure with unprecedented precision. FFGFT makes concrete predictions for the power spectrum without dark matter halo corrections.

New data (as of October 2026). Since the first edition, three results have appeared that bear directly on these questions.

Weak lensing around individual galaxies. In 2024 Mistele, McGaugh, Lelli, Schombert and Li used data from the Kilo-Degree Survey to follow the rotation curves of isolated galaxies through weak gravitational lensing far beyond their visible edges. The curves stay flat, reliably out to at least 300 kpc and without a visible decline out to about 1 Mpc, and they follow the baryonic Tully-Fisher relation. This is exactly the behaviour of the force law in the outer regime. A halo of particles would have to end somewhere; no end is visible so far.

Wide binary stars. Pairs of stars orbiting each other at separations of thousands of astronomical units lie below a_0 and are therefore a test in our own neighbourhood. The results contradict each other. In 2026 Chae and collaborators find, in 36 particularly well-measured pairs, gravity enhanced by about 60 %, a deviation from Newton at $4.9\,\sigma$. Saad and Ting show for the same pairs that the result depends on the assumption about the orbit sizes and is consistent with Newton under a different, equally reasonable assumption. For FFGFT this is open: what the force law looks like

for binaries in the gravitational field of the Milky Way has not yet been derived.

Direct detection. In 417 live days up to April 2025 the xenon detector LZ found no signal of the usual dark matter particles, but for the first time saw collisions of neutrinos from the solar core at $4.5\,\sigma$. In September 2026, however, the collaboration published a single event at about 248 keV that, by every criterion checked, looks like a heavy particle striking a xenon nucleus, against an expected background of one hundredth of an event. The significance is $2.6\,\sigma$, and according to the collaboration the analysis is not blind. Before further data are analysed, FFGFT has stated (Doc. 396): no accumulation of such events, no annual modulation peaking in early June, no independent confirmation in a second detector. A confirmed accumulation with a June maximum would refute the reading of dark matter as a regime. The other parts of FFGFT, from the particle sector to gravitation in the solar system, do not depend on it.

Chapter 9

An Honest Assessment

FFGFT explains galaxy rotation curves without dark matter. It links the acceleration scale to the Hubble constant, $a_0 = cH_0/(2\pi)$, without any further parameter. It has not yet passed the Bullet Cluster test; that test is open (Chapter 6). It explains why dark energy is not a physical object.

This is not a complete theory of gravity. Open questions remain:

The K3 forward proof — the mathematical derivation of the geometry from which the modified force law follows is structurally motivated but not yet fully proved.

14 % deviation at a_0: The derived value 1.03×10^{-10} m/s^2 lies about 14 % below the empirical value. The authoritative measurement from 153 galaxies gives 1.20 ± 0.02 (random) ± 0.24 (systematic) $\times 10^{-10}$ m/s^2 (McGaugh, Lelli, Schombert 2016). The FFGFT value thus lies within the systematic uncertainty, which comes from the mass-to-light ratio of the stars, but clearly outside the random one. The point is open.

Bullet Cluster and cluster physics: With the observed masses of gas and stars, the lensing peak in the Bullet Cluster lies at the gas, not at the galaxies. This test is open (Chapter 6, Doc. 308).

What is clear: assuming that 95 % of the universe consists of unknown substances that have never been directly detected is not a smaller assumption than assuming that inertia becomes nonlinear at very small accelerations.

The difference: the former assumption is not falsifiable — one can always add more dark matter when an observation fails to fit.

The latter has a precise, distinguished scale a_0 that follows from the structure of the theory and manifests in observations.

Dark matter is not wrong. It might exist. But it is not necessary. And a physics that manages without it deserves serious scrutiny.

Chapter 10

What Comes Next

The next generation of observations will decide.

The ESA Euclid telescope is surveying billions of galaxies out to redshifts of $z \sim 2$. The James Webb Space Telescope is showing us galaxies in an early universe that, according to the standard model, should not yet exist — but fit comfortably in the static FFGFT universe. DESI is measuring baryonic acoustic oscillations with record precision.

For FFGFT, concrete predictions remain open:

Achromatic redshift — no wavelength dependence across the entire observable range — is a core test. Current precision observations already set tight limits; FFGFT is so far compatible.

The mass distribution in galaxy clusters without dark matter halos: FFGFT cluster profiles differ subtly from NFW profiles of standard dark matter models. The resolution of future lensing surveys could distinguish them.

The baryonic Tully-Fisher relation holds in FFGFT exactly — not as a statistical trend with scatter, but as a geometric consequence. Deviations would be a direct indication of dark matter as a genuine substance.

Whether dark matter exists or is an illusion of the wrong physics is one of the greatest open questions in science. It is decidable. The theory that answers it will not be found through belief — but through predictions that either prove themselves or fail.

FFGFT stands ready to be tested.

Sources and Foundation Documents

This book is based on the FFGFT corpus in the GitHub repository https://github.com/jpascher/T0-Time-Mass-Duality. The following documents are the primary sources:

- **Doc. 308** — The cosmic sector of FFGFT (redshift, dark matter, Bullet Cluster, light deflection)
- **Doc. 267** — Cosmological degeneracy: ΛCDM and FFGFT as two readings of the same observations
- **Doc. 309** — The scale-anchor problem (H_0 derivation, SI anchor, asymmetry of the models)
- **Doc. 025** — T0 cosmology (CMB, ξ-field effects, static universe)
- **Doc. 004** — T0 model overview (fundamental relations, field geometries)
- **Doc. 012** — Gravitational constant from FFGFT
- **Doc. 392** — Where the representations of gravity part ways (readings of the redshift, DESI test)
- **Doc. 396** — Advance commitment on the LZ event at 248 keV

Recent measurements: McGaugh, Lelli, Schombert 2016, Phys. Rev. Lett. **117**, 201101 (RAR); Mistele et al. 2024, arXiv:2406.09685 (weak lensing); Chae et al. 2026, arXiv:2601.21728, and Saad, Ting 2026, arXiv:2603.11015 (wide binaries); LZ Collaboration 2025 and 2026 (direct detection).

All calculations in Chapters 4 and 5 were verified with the verification scripts `ffgft_308_p36_stufeA_a0_anker.py` and `ffgft_308_p36_stufeC1_bullet_cluster.py`.